REDUCEDNESS ASCENDS ALONG WEAKLY ETALE MORPHISMS

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 19080-19084 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 19086-19091, the complete author proof (4 body lines).

Editorial note. Navigation stubs: 2 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Eighth weakly etale-series card; mirrors L16162's structure with the reducedness dictionary.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Chapter lemma weakly-etale-characterize (card L16119); More on Algebra, lemma absolutely-flat-over-absolutely-flat (the ring-level ascent).

Source reference. Original lemma 094S: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 18935–18950. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 092I: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 19091–19091. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $f : X \to Y$ be a morphism of schemes. If Y is reduced and f weakly étale, then X is reduced.*

Proof. Via Lemma 094S this follows from the case of rings which is More on Algebra, Lemma 092I. □

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.